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Ankle Joint Proprioception and Postural Control in Basketball Players With Bilateral Ankle Sprains

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Background: Deficiencies in ankle proprioception and standing balance in basketball players with multiple ankle sprains have been reported in separate studies. However, the question of how ankle proprioceptive inputs and postural control in stance are related is still unclear.

Hypothesis: Ankle repositioning errors and the amount of postural sway in stance are increased in basketball players with multiple ankle sprains.

Study Design: Controlled laboratory study.

Methods: Twenty healthy male basketball players and 19 male basketball players who had suffered bilateral ankle sprains within the past 2 years were examined. Both groups were similar in age. Passive ankle joint repositioning errors at 5° of plantar flexion were used to test for ankle joint proprioception. The Sensory Organization Test was applied with dynamic posturography to assess postural sway angle under 6 sensory conditions.

Results: A significant increase in ankle repositioning errors was demonstrated in basketball players with bilateral ankle sprains ($P < .05$). The mean errors in the right and left ankles were increased from 1.0° (standard deviation, 0.4°) and 0.8° (standard deviation, 0.2°), respectively, in the healthy players to 1.4° (standard deviation, 0.7°) and 1.1° (standard deviation, 0.5°) in the injured group. A significant increase in the amount of postural sway in the injured subjects was also found in conditions 1, 2, and 5 of the Sensory Organization Test ($P < .05$). Furthermore, there were positive associations between averaged errors in repositioning both ankles and postural sway angles in conditions 1, 2, and 3 of the Sensory Organization Test ($r = 0.39-0.54$, $P < .05$).

Conclusions: Ankle repositioning errors and postural sway in stance increased in basketball players with multiple ankle sprains. A positive relationship was found between these 2 variables.

Clinical Relevance: Such findings highlight the need for the rehabilitation of patients with multiple ankle sprains to include proprioceptive and balance training.

Keywords: proprioception; postural sway; multiple ankle sprains; basketball players

Postural instability is a common disabling sequela in patients with a history of ankle sprains. About 40% of subjects with a history of such sprains suffer from a feeling of postural instability.⁹ This instability not only significantly disrupts an athlete's ability to participate but also curtails his or her ability to perform.²²

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Freeman et al⁹ postulated that during an ankle sprain, the nerve fibers of the mechanoreceptors located at the torn ligament could also be damaged. As a result, proprioceptive inputs from the ankle joint could be reduced. These authors further suggested that a reduction in the ankle proprioceptive inputs associated with an ankle sprain could be a predisposing factor for postural instability.⁹ Later studies indeed demonstrated that either ankle kinesthetic awareness or ankle repositioning was affected in subjects who had a history of ankle sprains.^{7,12,24} Garn and Newton¹² found a significant reduction in the detection of joint movement in the injured ankle when compared with the uninjured side. In 11 gymnasts with a history of more than 1 sprain of the same ankle, Forkin et al⁷ also reported a significant reduction in detecting movement of the injured ankle. In addition, Lentell et al²⁴

showed that the mean passive ankle inversion movement sense increased significantly (by 34%), from 3.2° in the uninjured ankles to 4.3° in the injured ankles.

Glencross and Thornton¹⁴ were the first group to quantify ankle proprioception by using a standard goniometer. They found greater errors in active repositioning of plantar flexion in the injured ankle, as compared with the uninjured ankle. It should be pointed out that a standard universal goniometer marked in 1-degree increments was probably not sufficiently sensitive to measure minor changes in the angles during a joint-repositioning test, especially when it was handheld rather than being firmly fixed to the joint. In addition, the speed of ankle plantar flexion was not standardized in their study. In later studies, a dynamometer was used to provide a constant speed of movement and to objectively measure the joint angle.^{18,19} A study by Konradsen et al¹⁹ assessed subjects with acutely sprained ankles. The total error in passive ankle repositioning to 3 positions (10° of eversion, 10° of inversion, and 20° of inversion) was found to be increased by 48% at 12 weeks after an acute sprain.¹⁹ The error in repositioning the ankle in inversion was also found to be significantly increased (by 34%) in a study by Jerosch and colleagues.¹⁸ However, Wright et al³⁹ pointed out that an increase in ankle plantar flexion, but not subtalar joint angle at touch-down, caused more occurrences of ankle sprains. Hence, the assessment of ankle proprioception during plantar flexion with the use of an accurate instrument and a reliable paradigm, as we have performed in the present study, was much needed.

Apart from the decrease in joint proprioception, postural sway in stance has been shown to be increased, under both eyes-open and eyes-closed conditions, in active athletes who had ankle sprains. Freeman and his associates⁹ observed that about 39% of subjects who had ankle sprains were more unsteady in single-limb stances on the injured leg. Investigators in more recent studies have quantified the amplitude of body sway using a force plate. They found postural sway to be increased in single-limb stances on the leg with ankle sprains^{10,23} and also when subjects were in double-leg stance.³³ However, the activity level of the injured and healthy groups was either not mentioned or not controlled. To avoid intersubject variability, Goldie et al¹⁵ used a within-subject design to compare postural sway in single-limb stance and assumed that the performance on the uninjured leg was not being affected by the injured leg on the contralateral side.

In this connection, a decrease in postural control has been detected in subjects with a loss of joint proprioception, such as in patients with peripheral neuropathy.^{6,40} However, proprioceptive deficits, induced experimentally by an ischemic block of the lower leg¹⁶ or anesthesia of the ankle joint,²⁰ had no effect on postural control in stance. In other words, results from these anesthetic studies could not demonstrate the influence of ankle proprioception on balance control during unperturbed stance. Nashner et al³⁰ pointed out that the control of standing balance lies in the coordinated motor outputs of somatosensory (including joint proprioceptive) inputs interacting with visual and vestibular cues. Probably because of the multiplicity of these inputs, when only 1 type of sensory input is altered

or absent, as in subjects with unilateral loss of vestibular function, relatively normal postural control in stance can still be maintained.³ In other words, the significance of ankle proprioception in the regulation of postural control in stance awaits further investigation.

In addition, previous studies separately reported deficits in joint proprioception and standing balance in patients with a history of ankle sprains. There has hitherto been no study investigating the relationship between ankle proprioceptive deficits and postural control in the same subjects. To examine the effect of ankle proprioceptive inputs on postural control in a more functional context, it is also necessary to assess postural control during stance under more challenging sensory conditions. For example, the control of body sway should be examined when visual and proprioceptive inputs are absent or in conflict with each other. Such an approach ensures that the sensorimotor balance systems are being more challenged, so that any deficits in postural control when standing under different (as opposed to constant) sensory conditions can be revealed. Possible relationships between ankle proprioceptive inputs and postural control in stance could then be determined for the sensory condition that requires the control of body sway to rely more heavily on ankle proprioceptive inputs.

The Sensory Organization Test (SOT) using the EquiTest system (NeuroCom International Inc, Portland, Ore) provides 6 different sensory conditions, which puts the sensorimotor balance system to greater challenge when visual and proprioceptive inputs are being manipulated, as opposed to a simple static stance under unchanged sensory conditions.^{1,27} With this test, possible maladjustment in postural control under different sensory conditions can be revealed. Hence, the main aims of our study were 3-fold: (1) to investigate possible deficiencies in the accuracy of ankle proprioception in basketball players who had sustained bilateral multiple ankle sprains, (2) to investigate possible postural maladjustment in body sway under reduced or conflicting sensory conditions in players who had sustained bilateral multiple ankle sprains, and (3) to assess the relationship between ankle proprioception and body sway in stance under the different sensory conditions. Findings from this study will provide a scientific basis for formulating novel rehabilitation approaches to treat subjects with a history of multiple ankle sprains. We hypothesized that there would be an increase in ankle repositioning errors and postural sway in basketball players who had sustained bilateral ankle sprains, under conditions in which they had to rely more heavily on ankle proprioceptive input. In addition, we believed that there would be a positive correlation between errors in ankle repositioning and the amount of postural sway in stance under different sensory conditions.

MATERIALS AND METHODS

Subjects

Thirty-nine male basketball players between 19 and 26 years of age were recruited from the basketball teams of 7 local universities in Hong Kong. Twenty subjects were

healthy, with no injury in either ankle for a minimum of 2 years. Nineteen others had experienced bilateral multiple ankle sprains (MAS), with at least 2 sprains to each ankle within 2 years but more than 3 months before this study. The sprains had been severe enough to require medical intervention. All the subjects in both groups had no previous fractures or surgery to the lower limb or back, and none of them suffered from any other problems in the musculoskeletal system. The study was approved by the Ethics Committee of The Hong Kong Polytechnic University, and all the participants signed an informed consent form.

Experimental Procedures

Ankle joint repositioning test. One commonly used method for testing joint repositioning is the assigned joint position test.³⁵ During this test, the joint is either passively or actively moved to a target position that is memorized by the subject. It is then returned to the initial position. Thereafter, the joint is moved either passively or actively back to the target position. In this study, the passive ankle joint repositioning test was chosen to assess possible proprioceptive deficits as consequences of multiple ankle sprains. In addition to the integrity of joint proprioception sense, success in reproducing the exact target position may be affected by the subject's memory, but this factor should not be a problem for the young and active subjects participating in the present study. Another consideration is that Konradson et al¹⁹ found passive but not active joint repositioning to be significantly affected when afferent information from the joint was suppressed. This result could be due to the fact that active joint repositioning could have been assisted by the "corollary" discharges at the cortical level when active voluntary movements are produced by a subject.²¹ In any case, owing also to the possible weakness of the ankle muscles consequent to ankle sprains, passive repositioning of the ankle joint was selected in our study so as to better delineate, by avoiding active muscle contraction, possible proprioception deficits in individuals with a history of ankle sprains.

Errors in joint repositioning were assessed in both ankles with the setup shown in Figure 1. Subjects lay prone with the tested foot padded with orthopaedic foam to reduce sensory input from the skin. The ankle was fixed in a neutral position of 90° of ankle dorsiflexion, with the forefoot strapped to the footplate of a Cybex Norm dynamometer (Cybex International Inc, Ronkonkoma, NY). The latter was used to provide ankle displacement at a constant speed. An electrogoniometer (Type XM180, Penny and Giles Biometric Ltd, Blackwood, Gwent, UK) was attached over the ankle joint, with its axes along the lateral aspect of the foot and the lower leg, to record the amplitude of ankle displacement in the sagittal plane at a sampling rate of 1000 Hz. The ankle was passively moved to 5° of plantar flexion at a peak velocity of 1 deg/s. It was held for 2 seconds once the target position was reached. The ankle was then returned to the neutral position and was again moved passively toward the target position at the same speed of 1 deg/s. When the subject perceived that



Figure 1. Experimental setup for the passive ankle joint repositioning test. Subjects lay prone, with the ankle in the neutral position of 90° of dorsiflexion and the forefoot strapped to the footplate of a Cybex Norm dynamometer. The foot was moved at a constant speed of 1 to 5 deg/s of plantar flexion.

the ankle had regained the previous target position, he produced an audio signal by pressing on a button held between his thumb and index finger. The initiation of this thumb action was captured by recording the onset of electromyographic (EMG) activity in the flexor pollicis brevis. The EMG signals were recorded through stainless-steel surface electrodes (NeuroCom International Inc) sampled at a rate of 1000 Hz, filtered with a bandwidth of 10 to 500 Hz, amplified with a gain of 4048, and stored for offline analysis (Dataq Instruments Inc, Akron, Ohio).

Sensory Organization Test. After the passive ankle joint repositioning test, the postural sway of each subject was assessed in stance under reduced or conflicting sensory conditions using the SOT.^{1,27,29} The subjects' task was to stand with as little sway as possible for 20 seconds under each of the 6 sensory conditions on the platform of a Smart EquiTest system (NeuroCom International Inc) (Figure 2). In conditions 1 and 2, subjects stood on a stable platform facing a fixed visual surround, with eyes open (condition 1) and closed (condition 2). In condition 3, subjects stood on a stable platform with a sway-referenced visual surround that was moved in conjunction to their body sway and that delivered "conflicting" visual information. In other words, subjects had to rely more heavily on ankle proprioception to maintain balance when the visual input was either absent (condition 2) or distorted (condition 3). In condition 4, subjects stood with their eyes open on a sway-referenced platform that was moved in conjunction to their body sway. They received inaccurate or misleading information from the foot support surface and had to rely more heavily on the visual and vestibular systems for balance control. In condition 5, subjects stood on the sway-referenced platform with their eyes closed; in condition 6, both the plat-



Condition	1	2	3	4	5	6
Vision	Normal	Absent	SR	Normal	Absent	SR
Platform	Fixed	Fixed	Fixed	SR	SR	SR

Figure 2. Sensory Organization Test. Subjects stood on a movable platform inside a movable visual surround and were instructed to stand as steady as possible under 6 different sensory conditions. SR, sway-referenced.

form and the visual surround were sway-referenced. Under these final 2 conditions, subjects had to rely more heavily on their vestibular inputs to coordinate motor outputs for maintaining their balance in stance. For each condition, 3 trials were performed for averaging purposes.

Data Recording and Analysis

Ankle joint repositioning test. The error in passive repositioning of the ankle from neutral to 5° of plantar flexion was calculated as the difference between the angles of the target position and the achieved reposition. The achieved reposition was determined when an onset of muscle activity was detected at the thumb (mainly the flexor pollicis brevis) muscle. The EMG onset was defined as the time when the rectified EMG signal exceeded the mean background level plus 3 standard deviations for 25 milliseconds.⁵ The background level was determined by averaging the EMG signals for 50 milliseconds before moving the ankle. Absolute error (without regard to the direction of the error) in ankle repositioning was used because it has been found to be the most accurate, repeatable, and precise means of expressing ankle proprioception sense, when compared with real error and error variation in a group of 20 ankle-healthy subjects.¹⁹

Sensory Organization Test. The sway angle quantified the anterior-posterior excursion of a subject’s sway at the center of mass during a 20-second trial. This excursion was calculated from the vertical forces collected by 2 dual force plates of the Smart EquiTest system, with the assumption that the center of mass was at 55% of the subject’s height.²⁹

Statistical Analysis

The Statistical Package for Social Sciences, version 10 (SPSS Inc, Chicago, Ill) was used for data analysis. Unpaired *t* tests were used to compare the age, height, and weight between the 2 groups. Repeated-measures analysis of variance (ANOVA) was then used to assess the errors in ankle joint repositioning, using group (MAS and healthy subjects) as a between-subject factor and leg (right and left) as a within-subject factor. Post-hoc analysis was conducted to assess the effect of leg and group factors on ankle joint repositioning errors. A multivariate ANOVA was used to compare the amount of body sway between the 2 groups because there was 1 independent variable—subject group (MAS and healthy subjects), and 6 dependent variables—sway angle under the 6 sensory conditions of the SOT. Multivariate regression analysis was conducted to establish the relationship between the averaged errors in passive joint repositioning of both ankles and the sway angle under the 6 sensory conditions. In all statistical analyses, a significance level (α) of .05 was chosen.

RESULTS

Table 1 shows the demographic data of the 2 subject groups. The MAS and healthy subjects had similar ages and weights, but the MAS group was, on average, slightly but significantly taller than the healthy group (by 0.04 m; $P < .05$). All subjects had participated in basketball competition for more than 5 years. In addition, the time since the most recent ankle sprain was 6.0 months (standard deviation [SD], 3.4 months) for the MAS subjects. All of them had sought medical intervention, but none had received rehabilitation after their injuries. Thirteen of the 19 subjects (63%) in the MAS group had feelings of postural instability, especially during landing from jumps.

Test-Retest Reliability

Ankle joint repositioning test. Seven male subjects participated in the pilot study examining data reproducibility of the passive ankle joint repositioning test, from neutral to 5° of plantar flexion at a peak speed of 1 deg/s. They were university basketball team players aged between 20 and 22 years. Five of them had experienced more than 3 ankle inversion sprains, with the most recent sprain being more than 3 months before the date of assessment. The 2 healthy players had no history of ankle sprain for the previous 2 years. All of them were test subjects in the main study.

Results from the pilot study indicated that the errors in passive joint repositioning of the ankle under our experi-

TABLE 1
Demographic Data of the Subjects

	Healthy (n = 20)		Multiple Ankle Sprains (n = 19)	
	Mean	SD	Mean	SD
Age, y	20.9	1.6	20.4	1.0
Weight, kg	68.5	8.7	70.6	8.0
Height, m	1.8	0.0	1.8	0.1 ^a
Competition, y	5.2	1.5	5.6	2.7
Time since last ankle sprain, mo			6.0	3.4

^aDenotes significant difference at $P < .05$ using unpaired t test.

mental paradigm had good reproducibility in the young basketball players. The intraclass correlation coefficients based on the 2-way mixed ANOVA model were 0.84 for both ankles, and the between-day variations were less than 3.5% and 2.5% on the right and the left ankles, respectively. In other words, the protocol that we adopted for the measurement of the errors in ankle repositioning to 5° of plantar flexion was repeatable and consistent.

Sensory Organization Test. Sixteen male basketball players were recruited by convenience for this pilot study examining the reproducibility of the sway angle in stance using dynamic posturography with the Smart EquiTest system to assess the control of their standing balance under reduced or conflicting sensory conditions. The healthy group consisted of 11 players from 20 to 23 years of age, and the injured group consisted of 5 subjects from 20 to 22 years of age. Seven of these subjects were test subjects in the main study, and 9 additional subjects were recruited for this reliability study.

Good test-retest repeatability was found in the averaged sway angle in each of the 6 sensory conditions. The intraclass correlation coefficients based on the 2-way mixed ANOVA model were 0.73, 0.89, 0.84, 0.89, 0.75, and 0.88 for conditions 1 to 6, respectively. Hence, the standing balance test using the Smart EquiTest system produced repeatable and consistent body sway angle in active male basketball players.

Ankle Joint Repositioning Test

Thirty-nine basketball players were recruited, but only 38 players participated in the passive ankle joint repositioning test. One subject from the healthy group was excluded because he was not able to stay until the test was finished, thus making 19 in each group. Because of technical error, there are 2 missing data points from the right ankle of the healthy subjects and 1 missing data point from the left ankle of an injured subject. Analyses were only made on the healthy and injured players who had complete data from both ankles ($n = 17$ in the healthy group and $n = 18$ in the injured group). Averaged errors in passive ankle repositioning in the 17 healthy players were 1.0° (SD, 0.4°)

TABLE 2
Mean Sway Angle (Degrees) Under the 6 Sensory
Conditions of the Sensory Organization Test

	Healthy (n = 20)		Multiple Ankle Sprains (n = 19)	
	Mean	SD	Mean	SD
Condition 1	0.7	0.1	0.8	0.2 ^a
Condition 2	0.9	0.2	1.1	0.3 ^a
Condition 3	1.0	0.4	1.1	0.3
Condition 4	2.2	1.2	2.0	0.9
Condition 5	4.5	1.3	5.7	1.4 ^a
Condition 6	4.6	1.6	5.6	1.8

^aDenotes significant difference at $P < .05$ using unpaired t tests, after multivariate tests showed an overall statistically significant difference at $P < .05$.

and 0.8° (SD, 0.2°) in the right and left ankles, respectively. No significant difference (post-hoc paired t test, $P > .05$) was found between the right and left passive ankle repositioning errors in these 17 healthy players.

When compared with the healthy group, the MAS group had increases in ankle joint repositioning errors at 5° of ankle plantar flexion, which were shown to be significant using repeated-measures ANOVA ($P < .05$). Post-hoc analysis using unpaired t tests was therefore conducted, and the errors in repositioning of the right and the left ankles were found to be significantly increased from 1.0° (SD, 0.4°) and 0.8° (SD, 0.2°), respectively, in the healthy players to 1.4° (SD, 0.7°) and 1.1° (SD, 0.5°) in the injured group. This finding represented an increase of 40.0% and 37.5% for the right and left legs, respectively.

Sensory Organization Test

A total of 39 (20 healthy and 19 injured) players participated in the SOT. Table 2 presents the mean postural sway angle obtained from the 6 sensory conditions of the SOT for the 2 groups. Generally, the amount of sway angle was greater in the injured group than in the healthy group. Multiple ANOVAs demonstrated an overall significant difference in the amount of sway angle between the 2 groups ($n = 39$, $P < .05$). Post-hoc tests indicated that the MAS group had sway angles significantly greater than those of the healthy group in conditions 1, 2, and 5 ($P < .05$). In contrast, the differences were insignificant in conditions 3, 4, and 6.

Correlation of Absolute Ankle Angle Errors With Body Sway Angle

To delineate the relationship between ankle joint repositioning errors and the sway angle under the different sensory conditions, a multivariate regression analysis was applied to the available data from the 35 subjects who were assessed using both the passive ankle joint repositioning test and the SOT. Table 3 and Figure 3 show that the averaged errors in passive repositioning of both ankles

TABLE 3
Correlation Between Averaged Errors in Passive Ankle Repositioning and Sway Angles in the 6 Sensory Conditions of the Sensory Organization Test

	Correlation Coefficient (<i>r</i>)	<i>R</i> ²	<i>P</i>
Condition 1	0.44	0.19	.007 ^a
Condition 2	0.54	0.29	.001 ^a
Condition 3	0.39	0.15	.020 ^a
Condition 4	−0.04		.834
Condition 5	0.016		.926
Condition 6	−0.194		.264

^aDenotes significant difference at *P* < .05 using Pearson product-moment coefficient of correlation.

were significantly related to the sway angles obtained in conditions 1, 2, and 3 (*P* < .05). The plots in Figure 3 also indicate a positive relationship between the 2 variables, in that the larger the averaged errors in repositioning the ankle were, the greater the sway angle in the 3 conditions, with the correlation coefficient (*r*) being 0.44, 0.54, and 0.39 for conditions 1 to 3, respectively. In addition, the averaged ankle repositioning errors accounted for 29% (*R*² = 0.29, *P* < .05) of the sway angle in condition 2, but for only 19% and 15% of the sway angle in conditions 1 and 3 (*R*² = 0.19 and 0.15 for conditions 1 and 3, respectively, *P* < .05) (Table 3).

DISCUSSION

Ankle Joint Proprioception in Healthy and Injured Players

As previously mentioned, both joint position and movement sense have been found to be reduced in players with sprained ankles when compared with healthy players.^{12,14,19,24} In this connection, Robbins and Waked³⁴ pointed out that an inability to accurately position the foot before landing could be one of the factors contributing to an increased susceptibility to subsequent ankle sprains. Hence, quantification of errors in ankle joint repositioning was used in our present study to assess the proprioceptive deficit in subjects with multiple ankle sprains. The error in active joint repositioning has been shown to be about 1° for normal ankles when the end-target positions were 15° to 30° of ankle plantar flexion, as measured with a standard manual goniometer.¹⁴ Interestingly, the error in passive repositioning was found in the present study to be similar in magnitude for the end-target position of 5° of ankle plantar flexion, being 1.0° (SD, 0.4°) and 0.8° (SD, 0.2°) in the right and left ankles, respectively, of our healthy basketball players, when assessed with an electronic goniometer. In addition, there were no differences between the dominant and nondominant legs in passive ankle joint repositioning errors in these healthy players, a finding that concurs with the Glencross and Thornton¹⁴ study of a group of young adults.

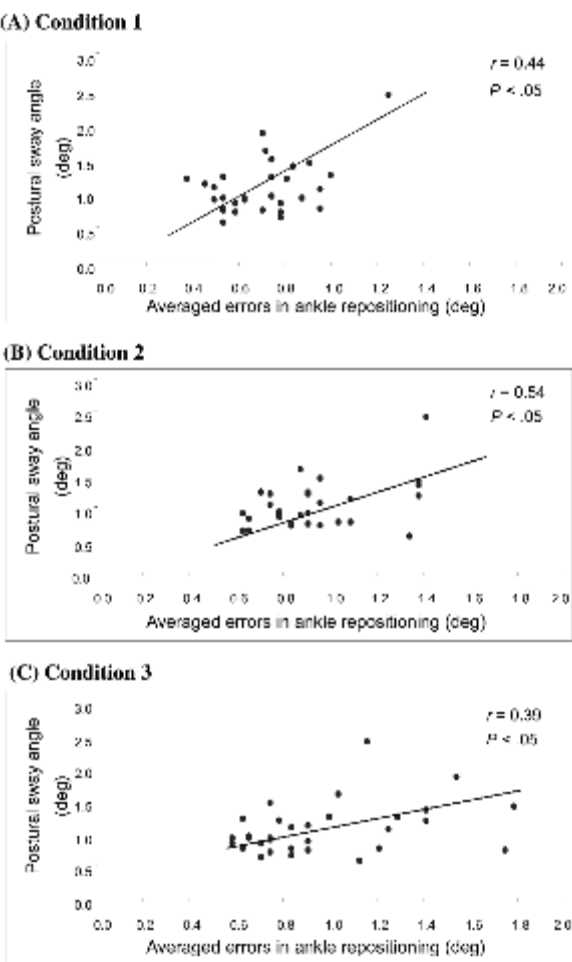


Figure 3. Regression lines correlating the averaged errors in passive repositioning of the ankle to 5° of plantar flexion with the sway angle obtained under (A) condition 1, (B) condition 2, and (C) condition 3.

In basketball players who had sustained bilateral MAS, the results from our present study provided concrete evidence that errors in passive joint repositioning at 5° of ankle plantar flexion were significantly increased when compared with those of healthy players. The percentage increase was 40.0% and 37.5% in the right and left ankles, respectively, or an average of 38.8%. These findings in the sagittal plane not only agreed with those from previous studies on the frontal plane but also yielded similar percentage changes to those reported for passive ankle inversion-eversion.¹⁸ Briefly, Jerosch and colleagues¹⁸ reported a deficit of 34% in subjects with chronic ankle sprains when compared with the contralateral healthy ankle during different degrees of inversion.

Effects of Multiple Ankle Sprains on Postural Sway

In this study, postural sway in stance was assessed using dynamic posturography under the 6 sensory conditions of

the SOT. Our results indicated that, generally speaking, postural sway in stance was significantly increased in basketball players with multiple ankle sprains when compared with healthy players. This finding was reflected by a significant increase in the sway angles when all conditions were considered in the multivariate analysis ($P < .05$) (Table 2). Post-hoc analysis further demonstrated that the main differences were found in conditions 1 and 2, in which the subjects had to rely on joint proprioception for balance with and without visual inputs, and in condition 5, in which coordinated motor output relied heavily on vestibular input in the presence of erroneous or absent signals from the ankle proprioceptive and visual systems (Table 2).

In conditions 1 and 2, postural sway was captured in double-leg stance on a fixed support platform with the eyes open in the first condition and closed in the second. Results from this study not only agreed with, but also strengthened, those of previous studies.^{15,23,33} However, both Perrin et al³³ and Leanderson et al²³ compared postural sway in a group of injured basketball players with another group of healthy, active subjects. Thus, they were comparing postural sway between 2 groups of subjects who were likely to have different activity levels. To avoid inter-subject (between-group) variability, Goldie et al¹⁵ used a within-subject design to compare postural sway in single-leg stance between the injured and uninjured legs of 24 active subjects. These authors assumed that the performance on the uninjured leg was not affected by the contralateral injured leg. In the present study, 39 subjects were recruited from 7 university basketball teams, that is, basketball players competing at an institutional level instead of just ad hoc basketball players. This requirement provided a sufficient number of injured and healthy players with similar activity levels to allow for controlled comparisons. The latter is substantiated by the fact that all the recruited players had participated in basketball competitions for more than 5 years and that all had practiced the sport for more than 2 hours a week. Our findings showed that postural control in stance was affected in players with bilateral MAS, even in the presence of visual inputs and even during double-leg stance, as in condition 1 of the SOT (Table 2).

It was interesting to find that the postural sway in condition 5 was significantly greater in the injured group than in the healthy group, being 5.7° (SD, 1.4°) and 4.5° (SD, 1.3°), respectively (Table 2). Under this condition, the subject had to maintain his balance on a sway-referenced platform with his eyes closed, in which he had to rely heavily on vestibular input to coordinate motor output for balance control in the presence of nulled ankle proprioceptive and blocked visual inputs.^{1,30} This finding thus suggested that some subjects in the MAS group might have suffered from vestibular malfunction in addition to proprioceptive deficits. In this connection, bilateral vestibular failure is often an undetected disorder of the peripheral labyrinths.⁴ A question thus emerges: Could there be undetected vestibular malfunction in the basketball players which predisposed them to MAS? In a previous study, we found

that 45% ($n = 5$ of 11) of MAS subjects suffered from vestibular malfunction, as demonstrated by abnormal nystagmus in the caloric test. Furthermore, subjects with both MAS and vestibular malfunction had greater postural sway in conditions 5 and 6 of the SOT, when compared with the postural sway of healthy or MAS-only subjects.¹¹ On the other hand, Weintraub³⁷ suggested that repetitive jarring trauma to the delicate otoliths in high-impact aerobics could be a cause of vestibular malfunction. Thus, vestibular malfunction might have occurred in the injured players through high-impact shock resulting from an uncoordinated landing from a jump while playing basketball. It is clear that more insight into possible vestibular contributions to postural control in the 2 study groups is needed.

Relationship Between Ankle Joint Proprioception and Balance Control in Stance

Even though the MAS group had greater postural sway when compared with their healthy counterparts, the differences might be due to several factors, such as muscle weakness or ankle proprioceptive deficits as a result of repeated injuries to the ankle. This was the first study to correlate the extent of ankle proprioceptive deficit with postural sway in stance under reduced or conflicting sensory conditions. Our results revealed that there was a positive correlation between averaged errors in passive ankle joint repositioning to 5° of plantar flexion and the sway angle obtained under 3 different sensory conditions when subjects stood on a stable platform (conditions 1 to 3) (Figure 3). More specifically, the averaged errors in passive repositioning of the ankle on both sides were positively related to the amount of postural sway for conditions 1, 2, and 3, that is, the larger the errors in ankle repositioning, the greater the postural sway for these 3 conditions (Figure 3).

Results from our study indicated a positive relationship between averaged errors in passive ankle joint repositioning and amount of postural sway in stance. In addition, the positive relationship between these 2 variables only existed when subjects stood on a fixed (conditions 1 to 3) but not on a movable surface (conditions 4 to 6). Nashner²⁸ suggested that the contribution of different sensory inputs to postural control in stance depends on the task and the stability of the supporting surface. This study proposed that under normal circumstances, somatosensory (including ankle proprioceptive) input is the most important one in maintaining balance when the supporting surface is fixed (conditions 1 to 3 of the SOT). A later study by Mergner and Rosemeier,²⁶ using a model of vestibular-somatosensory interaction, supported the above notion that healthy subjects relied heavily on somatosensory (including ankle proprioceptive) inputs for balancing on a stable surface. When there were conflicting inputs between the somatosensory and the visual systems, as when standing on a stable surface with a moving visual surround, the vestibular system would then provide an internal reference frame for the integration of the sensory inputs (ie, condition 3 of the SOT).³⁰

In addition, the highest association between ankle proprioception and postural sway was found in condition 2 ($R^2 = 0.29$) (Table 3), when compared with conditions 1 and 3 ($R^2 = 0.19$ and 0.15 , respectively). In condition 2, subjects had to maintain their balance on a stable platform with their eyes closed, and somatosensory (including ankle proprioceptive) information provided the main source of sensory inputs for balance control in the absence of visual input and vestibular stimulation. Results from this study indicated that ankle proprioception alone accounted for 29% of the sway angle in condition 2. In other words, somatosensory inputs from other joints, muscles, and cutaneous stimulation could have also contributed to the control of balance in this condition. Despite the addition of visual input in condition 1, ankle proprioception still accounted for 19% of the sway angle. This finding could confirm the contribution of ankle proprioception to postural stability even in the presence of visual input.

Clinical Significance

Ankle sprain is one of the most common sports injuries, particularly in basketball. About 40% of injured athletes have a feeling of postural instability, which may predispose them to further injuries. In this connection, a study by Payne et al³² has shown that ankle joint proprioception deficits can be used to predict ankle joint injury in college basketball players. A more recent study by McGuine et al²⁵ further proved that preseason balance measurement in terms of postural sway per second, as measured by a system similar to the one used in the present study, served as a predictor of ankle sprain susceptibility in high school basketball players. In their study, the rate of postural sway per second increased from 1.7 deg/s in the group with no ankle injuries to 2.0 deg/s in the group with ankle injuries. In the present study, a significant increase in both ankle repositioning errors and the amount of postural sway in stance was detected in the injured basketball players with bilateral ankle sprains. Such findings suggest that the injured players are more susceptible to further ankle sprains, even though they are still participating in high-level competition. It is worth noting that these injured players did not receive any rehabilitation after their initial and subsequent injuries.

Proprioception and coordination have been regarded as important components in rehabilitating ankle sprains, in particular for individuals with functional ankle instability.^{8,13,17,31,36,38,41} Tropp et al³⁶ were the first to examine the effects of an ankle disk training program in 65 male soccer players with previous ankle sprains. Results from their study indicated an 80% decrease in the frequency of repeat sprains over a 6-month period in the exercise group when compared with the nonexercise group. The effects of wobble board training were also demonstrated in a prospective study conducted by Wester et al³⁸ in patients with ankle sprains. Findings from their study indicated a 50% reduction in reinjury rates in the exercise group when compared with controls. In this connection, Gauffin and colleagues¹³ demonstrated that an 8-week postural training program

using a wobble board caused a decrease in postural sway in single-leg stance, for both the trained and the untrained leg, in patients with functional instability of the ankle joint. Similar findings were reported by Bernier and Perrin.² After 6 weeks of balance and coordination exercises, subjects with functional ankle instability showed a significant improvement in their postural sway when compared with the nonexercise and sham groups. Nevertheless, there were insignificant differences for joint position sense among the 3 groups. The authors suggested that the insignificant findings in joint position sense for the exercise group could be related to the lack of specificity of the training. Indeed, both ankle disk and wobble board training would probably enhance the awareness of posture and position of the body in space, with the aim of maintaining equilibrium without changing the base of support. However, the ability to accurately position the foot has not specifically been trained. Specific ankle proprioceptive exercises could include accurate placement of the ankle or foot at a particular position in space. An improvement in ankle proprioception may thereby enhance standing balance. Such a suggestion arises because findings from this study indicated a positive association between ankle repositioning and postural sway in stance. Determining the effectiveness of a combined program of specific proprioceptive exercises and balance training awaits further research.

CONCLUSION

We concluded that there were increases in passive ankle joint repositioning errors and the amount of postural sway in stance in male university basketball players who had sustained MAS, when compared with healthy players similar in age, gender, and activity level. A positive correlation was also found between the averaged errors in passive ankle repositioning and the amount of postural sway in stance on a stable platform, particularly with eyes closed. This result demonstrated that the increase in postural sway during stance on a stable platform in basketball players with bilateral MAS might be related to an increase in errors in passive ankle repositioning. Such findings form a scientific basis that highlights the need for any rehabilitation program for subjects with multiple ankle sprains to include both proprioceptive and balance training.

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